VIVEKANAND EDUCATION SOCIETY'S
INSTITUTE OF TECHNOLOGY

Experiment 9
Mini Project Report

APPLICATION OF NATURE-INSPIRED COMPUTING TO A REAL-WORLD OPTIMIZATION PROBLEM

VEHICLE ROUTE OPTIMIZATION USING ANT COLONY OPTIMIZATION

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CLASS	D16 AD A/B
SUBJECT	NIA Lab
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Experiment 9 Mini Project | November 2026

A reproducible synthetic delivery-route study using Ant Colony Optimization, a nearest-neighbor baseline, a random-tour baseline, exact enumeration, and an independent route audit. All locations and distances in this demonstration are synthetic.

1. Project Summary and Problem Formulation

Selected topic: Project 5, Vehicle Route Optimization Using Ant Colony Optimization (ACO). The task finds a short closed tour that starts at a depot, visits every delivery stop once, and returns to the depot. The demonstration uses Euclidean distance on a small synthetic coordinate map.

Study design

Item	Specification
Locations	10 total: one depot and nine synthetic stops
Distance model	Euclidean distance in illustrative kilometre units
Route structure	Start at depot, visit each stop once, return to depot
ACO ants	48
ACO iterations	160
Parameters	alpha=1.0; beta=4.0; evaporation=0.42; deposit=100
Seed	141
Validation	Exhaustive enumeration of 9! possible tours
Primary metric	Closed-tour distance in km

Decision variables and constraints

A solution is an ordered permutation of all nine delivery stops between two depot entries. Every stop must appear exactly once; the depot must be both the first and last node; all route legs use nonnegative distances from the symmetric matrix. The report recomputes total distance from each consecutive pair of stops. Real road networks, traffic, vehicle capacities, time windows, driver hours, route restrictions, and live addresses are out of scope.

Privacy and use guardrails

The map is fully synthetic and uses no customer names, addresses, or personal delivery records. The application must not be populated with private addresses without authorization and suitable privacy controls. Distance is straight-line demonstration distance, not an operational travel estimate. Exact enumeration provides a lower bound and validity reference only for this ten-node instance.

2. ACO Method and Implementation

Each ant builds a tour one stop at a time. The transition weight combines the pheromone trail on an edge with inverse distance. After a tour is complete, 2-opt reverses sub-paths whenever that shortens the closed tour. Pheromone evaporates, then the iteration-best and global-best tours deposit pheromone on their edges.

Transition and pheromone rules

```
probability(i -> j) proportional to pheromone(i,j)^alpha * (1/distance(i,j))^beta
pheromone <- (1 - evaporation) * pheromone
iteration-best and global-best tours deposit Q / route_length
```

Pseudocode

```
initialize symmetric pheromone and inverse-distance heuristic
for each iteration:
    for each ant:
        grow a closed tour using roulette-wheel transitions
        improve the tour with deterministic 2-opt
    evaporate pheromone
    reinforce iteration-best and global-best edges
independently verify route structure and recompute every leg
```

Validation checks

Check	Expected	Observed
Depot starts and ends route	True	True
Every delivery stop appears once	9 unique stops	True
All stops included	All 9 IDs	True
Optimizer equals recomputed legs	Equal to 1e-9 km	True
ACO route is valid	True	True
Gap to exact optimum	Reported transparently	0.000000%

Implementation excerpt

```
for ant in range(ants):
    route = [depot]
    while unvisited_stops:
        next_stop = roulette_wheel(pheromone, inverse_distance)
        route.append(next_stop)
    route.append(depot)
    route = two_opt(route, distances)
    update_best_route(route)

assert independent_audit(route).route_is_valid
```

3. Benchmark Results and Best Route

All methods use the same ten-location distance matrix. The random baseline reports the mean and best of 500 seeded tours; nearest neighbor is deterministic; ACO uses 48 ants for 160 iterations with 2-opt; exact enumeration checks every one of the 9! possible delivery-stop permutations.

Benchmark comparison

Method	Distance (km)	Runtime (s)	Notes
Random (500 tours)	Mean 53.455; best 39.457	0.0542	500 seeded tours
Nearest neighbor	43.401	0.0442	Greedy nearest unvisited stop
ACO + 2-opt	31.154	5.2977	ACO candidate and 2-opt refinement
Exact enumeration	31.154	2.5082	Global optimum for this small instance

ACO + 2-opt returns 31.154 km. Exact enumeration also returns 31.154 km, so the best route on this synthetic instance matches the enumerated optimum with a 0.000000% gap. This does not establish that ACO always finds an optimum; the exact check is feasible here because there are only nine delivery stops.

Optimal closed route

VESIT Depot -> South Exchange -> Garden Square -> Lakeview Hub -> North Arcade -> East Works -> Riverside Depot -> Hillcrest Store -> Cedar Market -> Harbour Point -> VESIT Depot

Leg	From	To	Distance (km)	Cumulative (km)
1	VESIT Depot	South Exchange	3.245	3.245
2	South Exchange	Garden Square	3.581	6.825
3	Garden Square	Lakeview Hub	2.828	9.654
4	Lakeview Hub	North Arcade	2.404	12.058
5	North Arcade	East Works	3.000	15.058
6	East Works	Riverside Depot	3.396	18.454
7	Riverside Depot	Hillcrest Store	4.428	22.882
8	Hillcrest Store	Cedar Market	2.802	25.684
9	Cedar Market	Harbour Point	2.470	28.154
10	Harbour Point	VESIT Depot	3.000	31.154

Best route distance	Stops	Legs	Exact gap
31.154 km	9	10	0.000000%

4. Synthetic Map and Distance Tables

The coordinate locations are invented solely to make the optimization reproducible. The distance matrix is symmetric with a zero diagonal. Values are Euclidean units presented as illustrative kilometres; they are not road distances.

Location coordinates

ID	Location	X	Y
DEPOT	VESIT Depot	0.0	0.0
STOP-01	Harbour Point	2.4	1.8
STOP-02	Cedar Market	4.7	0.9
STOP-03	North Arcade	5.6	3.5
STOP-04	Lakeview Hub	3.9	5.2
STOP-05	Garden Square	1.1	4.8
STOP-06	East Works	7.4	5.9
STOP-07	Riverside Depot	8.2	2.6
STOP-08	Hillcrest Store	6.3	-1.4
STOP-09	South Exchange	-1.8	2.7

Pairwise distance matrix (km)

From / to	DEPOT	S01	S02	S03	S04	S05	S06	S07	S08	S09
DEPOT	0.0	3.0	4.8	6.6	6.5	4.9	9.5	8.6	6.5	3.2
S01	3.0	0.0	2.5	3.6	3.7	3.3	6.5	5.9	5.0	4.3
S02	4.8	2.5	0.0	2.8	4.4	5.3	5.7	3.9	2.8	6.7
S03	6.6	3.6	2.8	0.0	2.4	4.7	3.0	2.8	4.9	7.4
S04	6.5	3.7	4.4	2.4	0.0	2.8	3.6	5.0	7.0	6.2
S05	4.9	3.3	5.3	4.7	2.8	0.0	6.4	7.4	8.1	3.6
S06	9.5	6.5	5.7	3.0	3.6	6.4	0.0	3.4	7.4	9.7
S07	8.6	5.9	3.9	2.8	5.0	7.4	3.4	0.0	4.4	10.0
S08	6.5	5.0	2.8	4.9	7.0	8.1	7.4	4.4	0.0	9.1
S09	3.2	4.3	6.7	7.4	6.2	3.6	9.7	10.0	9.1	0.0

The ACO total is independently computed as the sum of the ten ordered route-leg distances. The complete distance matrix is included so that each result can be reproduced or recalculated without relying on a map service.

5. Colab Notebook and Reproducibility

The linked Colab notebook bootstraps the public project repository, runs the self-contained route optimization cells, prints the full benchmark and route tables, draws a black-and-white convergence chart, and exports CSV files.

Google Colab notebook

https://colab.research.google.com/drive/1GxqpVK-q_ZGQ7XSdVVrjEuHLYpR0fctJ

Notebook result tables and CSV exports

Output	Description
Benchmark comparison	Random 500-tour mean/best, nearest neighbor, ACO + 2-opt, exact optimum and runtime
Best route legs	Every ordered stop-to-stop leg and cumulative distance
Synthetic locations	Location IDs and X/Y coordinates
Distance matrix	All 100 symmetric pairwise distances
Independent audit	Route closure, one visit per stop, complete stop set, recomputed distance and exact gap
Convergence data	Best-so-far distance for each ACO iteration
CSV files	route_benchmark_results.csv, optimized_route_legs.csv, synthetic_locations.csv, pairwise_distance_matrix.csv, aco_convergence_summary.csv, route_guardrail_audit.csv

Reproduction steps

Open the notebook link and run all cells. The map and algorithm seed are fixed. The random baseline samples 500 tours; exact enumeration explores every permutation of nine stops. Runtime is measured per run and varies by hardware.

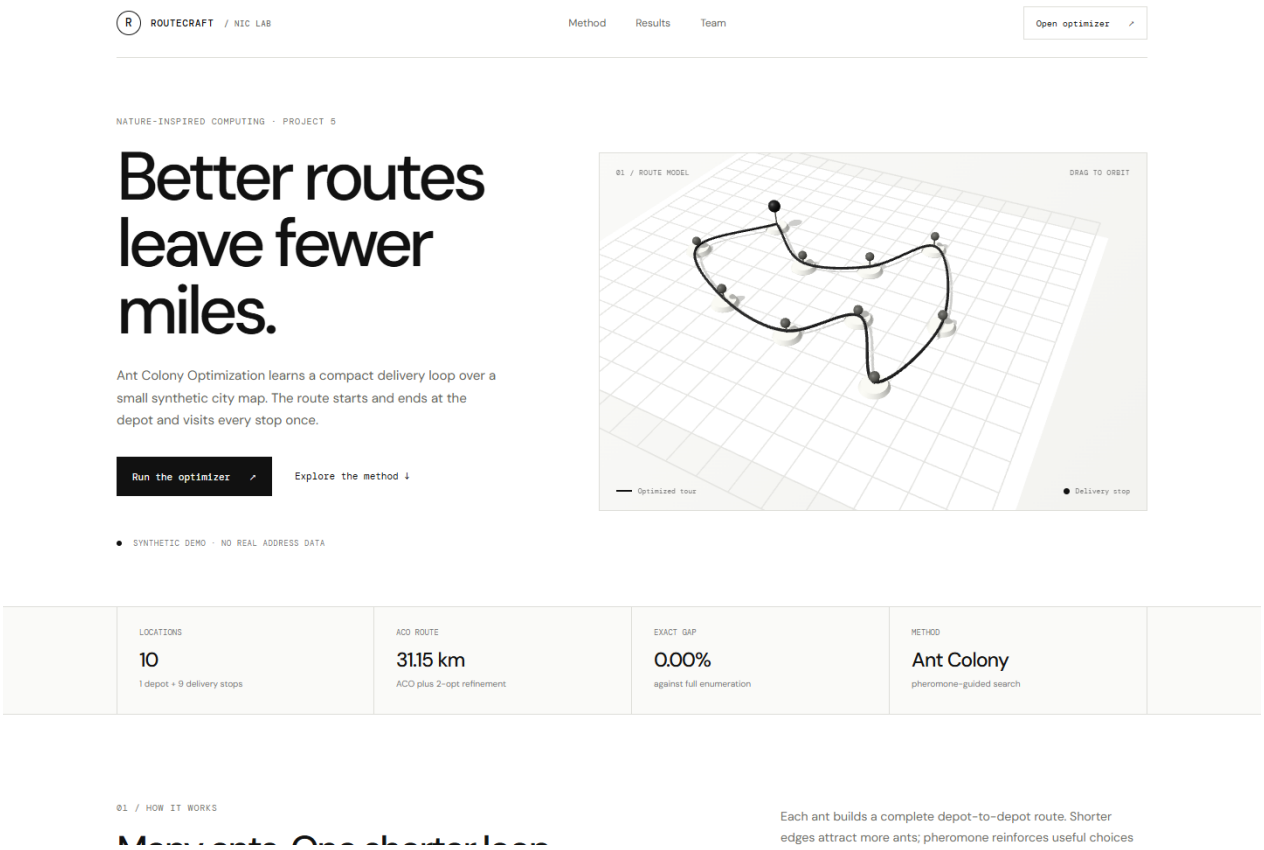
Scope limits

This synthetic benchmark does not model roads, traffic, travel-time variation, vehicle capacity, time windows, depot hours, loading limits, driver breaks, or legal routing restrictions. It is a class project demonstration, not dispatch advice. Do not infer operational route quality from straight-line distance.

```
route_distance = sum(distance[route[i]][route[i + 1]]
                     for i in range(len(route) - 1))
assert route[0] == route[-1] == depot
assert sorted(route[1:-1]) == all_delivery_stops
```

6. Website Evidence

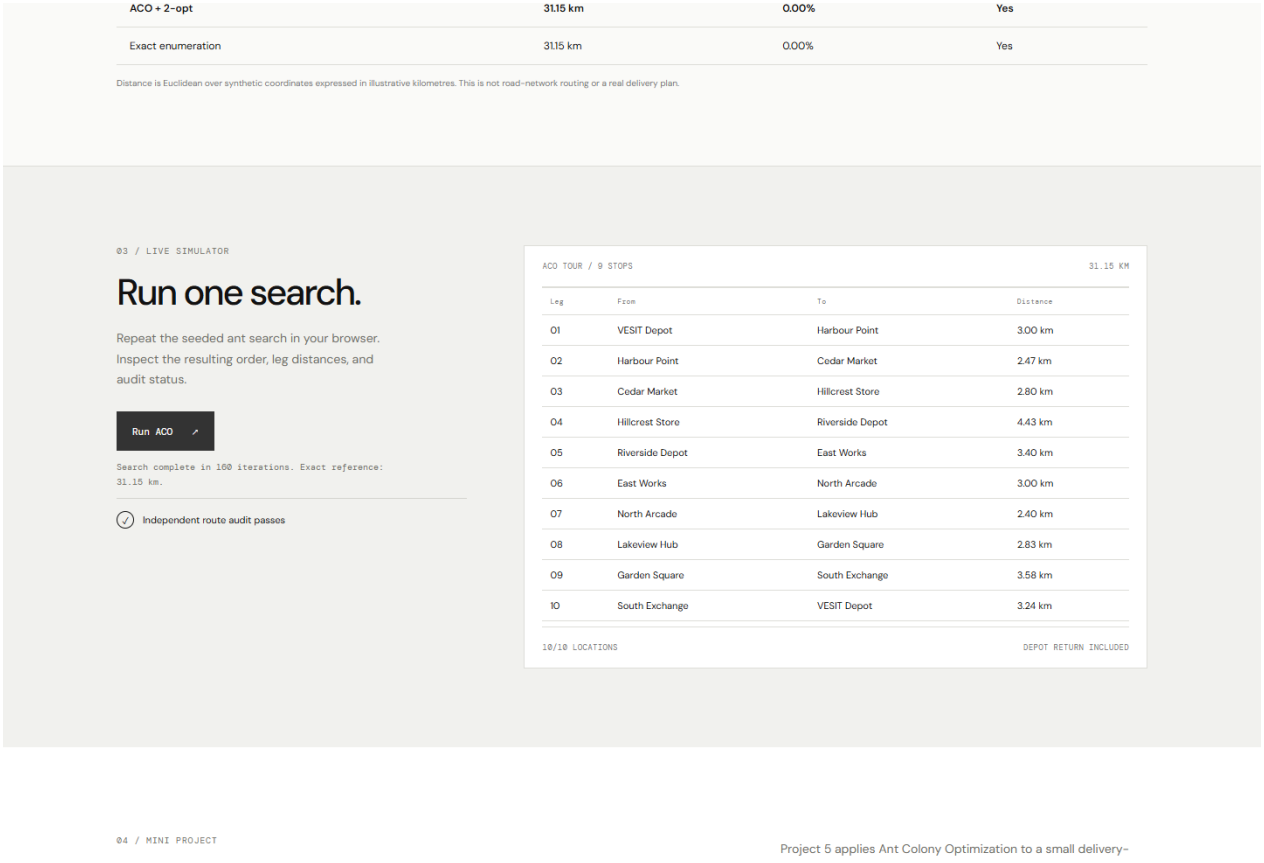
Landing page showing the monochrome 3D route network.



The website is designed with a white background, black text, grayscale 3D location pins, and an interactive route line. Drag the scene to rotate the model; the ACO control displays a computed route. The production website is deployed at https://nia-dist.vercel.app/ and was checked successfully with an HTTP 200 response and the expected Routecraft page title. The included screenshots show the same monochrome 3D website.

7. Website Results Evidence

Interactive ACO results table and route distance output.



ACO + 2-opt	31.15 km	0.00%	Yes
Exact enumeration	31.15 km	0.00%	Yes

Distance is Euclidean over synthetic coordinates expressed in illustrative kilometres. This is not road-network routing or a real delivery plan.

03 / LIVE SIMULATOR

Run one search.

Repeat the seeded ant search in your browser. Inspect the resulting order, leg distances, and audit status.

Run ACO ↗

Search complete in 100 iterations. Exact reference: 31.15 km.

✓ Independent route audit passes

ACO TOUR / 9 STOPS — 31.15 KM

Leg	From	To	Distance
01	VESIT Depot	Harbour Point	3.00 km
02	Harbour Point	Cedar Market	2.47 km
03	Cedar Market	Hillcrest Store	2.80 km
04	Hillcrest Store	Riverside Depot	4.43 km
05	Riverside Depot	East Works	3.40 km
06	East Works	North Arcade	3.00 km
07	North Arcade	Lakeview Hub	2.40 km
08	Lakeview Hub	Garden Square	2.83 km
09	Garden Square	South Exchange	3.58 km
10	South Exchange	VESIT Depot	3.24 km

10/10 LOCATIONS — DEPOT RETURN INCLUDED

04 / MINI PROJECT

Project 5 applies Ant Colony Optimization to a small delivery-

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References and Project Files

Supplied Experiment 9 mini-project brief: Project 5, Vehicle Route Optimization Using Ant Colony Optimization.

Project repository: https://github.com/Hardik182005/NIA
Production website: https://nia-dist.vercel.app/

Implementation prompt: PROJECT_PROMPT.md. Source implementation: scripts/route_optimizer.py. Notebook: notebooks/NIA_Vehicle_Route_ACO.ipynb. Website source: site/dist/.

Prepared for	NIA Lab | Experiment 9 Mini Project
Selected topic	Project 5: Vehicle Route Optimization Using ACO
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Data notice	Synthetic coordinates and distances only